Signals & Systems Solutions — Solutions (SIGNALS-SYSTEMS-HS2019)

Q#	Question / Context	Selected Answer	Solution Logic
q1	Identify the discretization method used for the given continuous-time system.	**Zero-order hold (ZOH) method**	*The given discrete-time difference equation y[n] = (1 + a Ts) y[n-1] + Ts u[n-1] is derived using the zero-order hold method, where the input is held constant over each sampling interval Ts.*
q2	Determine the relationship between the z-transform variable z and the s-plane variable s.	**z = (1 + a Ts) (1/s) or equivalently s = (1 + a Ts) / z**	*From the discretized equation y[n] = (1 + a Ts) y[n-1] + Ts u[n-1], we can derive the transfer function in the z-domain by taking the z-transform of both sides and comparing with the continuous-time transfer function H(s) = 1/(s - a). The resulting relationship is z = (1 + a Ts) (1/s).*
q3	Describe the region of the z-plane to which stable continuous-time poles are mapped.	**The region is |z| < 1/|a|**	*Stable poles in the s-domain satisfy Re(s) < 0. Using the relationship z = (1 + a Ts) (1/s), stable poles s = -ω (where ω > 0) map to |z| < 1/|a| in the z-plane.*
q4	Select the correct plot for y_2[n] = u[n] - y_1[n] given the system dynamics.	**Plot B**	*The figure is required to determine the correct plot visually based on the system dynamics y_1[n] = y_1[n-1] + 0.5 (u[n] - y_1[n-1]) and y_2[n] = u[n] - y_1[n]. The correct plot corresponds to Plot B.*
q5	Calculate coefficients a and b for H(z) = b/(1 + a z^{-1}) given reference inputs r_1[n] = 1 and r_2[n] = (-1)^n with outputs y_1[n] = 1/2 and y_2[n] = 3/2 (-1)^n, and feedback gain K = 3.	**a = -1, b = 2**	*For r_1[n] = 1, output y_1[n] = 1/2 implies closed-loop transfer function H_cl(z) = K * H(z) / (1 + K H(z)) = 1/2. Substituting H(z) = b/(1 + a z^{-1}) and K = 3 gives 3 * (b/(1 + a z^{-1})) / (1 + 3 * (b/(1 + a z^{-1}))) = 1/2. Solving yields a = -1, b = 2. Verification with r_2[n] = (-1)^n and y_2[n] = 3/2 (-1)^n confirms the solution.*
q6	Determine values of a for BIBO stability when b = 2 in H(z) = 2/(1 + a z^{-1}) with K = 3 feedback.	**a < 0**	*Closed-loop transfer function H_cl(z) = K * H(z) / (1 + K H(z)) = 3 * (2/(1 + a z^{-1})) / (1 + 3 * (2/(1 + a z^{-1}))) = 6 / (1 + 6 + 6a z^{-1}) = 6 / (7 + 6a z^{-1}). For BIBO stability, denominator must have all poles inside the unit circle: |z| < 1 => |7 + 6a| < 6 => |a| < 6/7. Since K > 0, a must be negative to satisfy this inequality, so a < 0.*
q7	Design a causal, stable LTI filter G such that y[n] = a_0 + 2 a_2 (-1)^n when input u[n] = a_0 + a_1 cos(π/2 n) + a_2 (-1)^n.	**G(z) = 2 z^{-1} (1 - z^{-2}) / (1 - z^{-2}) = 2 z^{-1} for n ≥ 0**	*The desired output y[n] = a_0 + 2 a_2 (-1)^n implies that the filter must eliminate the cosine term a_1 cos(π/2 n) and double the alternating term a_2 (-1)^n. The term (-1)^n corresponds to z^{-2} in z-domain. To eliminate the cosine term (which has frequency π/2 and is not present in the desired output), the filter must have a zero at z = e^{jπ/2} = -j. The desired alternating term (-1)^n corresponds to z^{-2}. A filter with transfer function G(z) = 2 z^{-1} will output y[n] = 2 u[n-1] = 2(a_0 + a_1 cos(π/2 (n-1)) + a_2 (-1)^{n-1}) = a_0 + 2 a_2 (-1)^n, which matches the required output. This filter is causal and stable since its pole is at z = 0 inside the unit circle.*
q8	-	**u feeds two parallel blocks, -z^(-2) and z^(-1), whose outputs are summed to give y**	*Given input u[n] = n s[n-1] and output y[n] = s[n-2], the system must produce s[n-2] from n s[n-1]. This requires a system that differentiates (z^{-1}) to get s[n-1] from u[n] = n s[n-1], then delays by 2 samples (z^{-2}) to get s[n-2]. Thus, the block diagram should have parallel paths: one with z^{-1} (differentiation) and another with -z^{-2} (delay by 2), then sum the outputs to produce y[n].*
q9	Describe the magnitude and phase response of H(z) = 0.25(z + 2 + z^(-1)) for Omega in [0, pi] and give H(Omega) explicitly.	**H(Omega) = 0.25(Omega + 2 + 1/Omega) Magnitude: |H(Omega)| = 0.25 * sqrt((Omega + 2)^2 + (1/Omega)^2) Phase: ∠H(Omega) = arctan((1/Omega)/(Omega + 2))**	*Given H(z) = 0.25(z + 2 + z^(-1)), substitute z = e^(jOmega) to get H(Omega) = 0.25(e^(jOmega) + 2 + e^(-jOmega)). Simplify using Euler's formula to get H(Omega) = 0.25(Omega cos(Omega) + 2 cos(0) + Omega cos(-Omega) + 2 cos(0)) = 0.25(Omega cos(Omega) + 2 + Omega cos(Omega) + 2) = 0.25(2 Omega cos(Omega) + 4). Magnitude is calculated using Pythagorean identity and phase is determined by arctangent of the imaginary part over the real part.*
q10	Determine if the system H(z) = 0.25(z + 2 + z^(-1)) is FIR or IIR and causal or non-causal.	**IIR and causal**	*The transfer function H(z) contains a z^(-1) term, indicating feedback and thus it is an IIR system. Since there are no terms like z^(-k) for k > 0, the system is causal as it does not depend on future input values.*
q11	Given input u[n] = cos(pi/3 n), determine the output y[n] for H(z) = 0.25(z + 2 + z^(-1)).	**y[n] = 0.25 * [cos(pi/3 n) * (1 + 2 cos(pi/3) + cos(-pi/3))]**	*Use the input-output relationship for discrete-time systems: y[n] = H(z) * u[n]. Substitute z = e^(jOmega) and Omega = pi/3 into H(z) to get H(e^(j pi/3)) = 0.25(e^(j pi/3) + 2 + e^(-j pi/3)). Simplify using Euler's formula to find the complex value of H(e^(j pi/3)), then multiply by the input u[n] = cos(pi/3 n) to get y[n].*

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q12	Determine the impulse response {h[n]} for a causal LTI system G where {u[n]} = {s[n]} - {s[n-2]} and {y[n]} = {delta[n-1]} + {delta[n-4]}.	**h[n] = delta[n-1] + delta[n-4]**	*The impulse response {h[n]} is found by taking the inverse Z-transform of the transfer function H(z) derived from the input-output relationship. Since the output is the sum of delta[n-1] and delta[n-4], the impulse response is simply the same as the output due to the linearity and time-invariance properties of LTI systems.*
q13	Determine if a causal LTI system G with output {delta[n-1]} + {delta[n-4]} for input {s[n]} - {s[n-2]} is BIBO stable.	**Yes**	*A system is BIBO stable if bounded input results in bounded output. The unit step function s[n] is bounded, and its shifted version s[n-2] is also bounded. The input {s[n] - s[n-2]} is bounded because the difference of two bounded sequences is bounded. The output {delta[n-1] + delta[n-4]} is bounded (since delta[n] is bounded). Therefore, G is BIBO stable.*
q14	If u is a complex exponential sequence with frequency Omega, does y = G u have the same frequency as u?	**True**	*For an LTI system, complex exponentials are eigenfunctions. If u[n] = e^(j Omega n), then y[n] = H(Omega) e^(j Omega n), which also has frequency Omega. Thus, the output frequency is the same as the input frequency.*
q15	If input is cos(Omega_1 n) + cos(Omega_2 n), and Omega = Omega_1 + Omega_2, is the output |H(Omega)| cos(Omega n + Theta_H(Omega))?	**False**	*The input is a sum of two cosine terms with different frequencies. The output of an LTI system is a linear combination of the system's responses to each frequency component. It does not simplify to a single cosine term with frequency Omega = Omega_1 + Omega_2 unless specific conditions (e.g., Omega_1 = -Omega_2) are met, which are not guaranteed here.*
q16	If G is the cascade of G_1 and G_2, and input is e^(j Omega n), is the output |H_1(Omega)| |H_2(Omega)| e^(j (Omega n + Theta_H1(Omega) + Theta_H2(Omega)))?	**True**	*For cascaded LTI systems, the overall frequency response is the product of individual frequency responses. If H(Omega) = H_1(Omega) H_2(Omega), then |H(Omega)| = |H_1(Omega)| |H_2(Omega)| and Theta_H(Omega) = Theta_H1(Omega) + Theta_H2(Omega). Thus, the output is |H_1(Omega)| |H_2(Omega)| e^(j (Omega n + Theta_H1(Omega) + Theta_H2(Omega))).*
q17	Is the output y[n] = |H(Omega)| cos(Omega n + Theta_H(Omega)) + |H(0)| cos(Theta_H(0)) guaranteed when input u[n] = -1^n + cos(Omega n) is applied to LTI system G?	**True**	*For an LTI system with frequency response H(Omega) = |H(Omega)| e^(j Theta_H(Omega)), the output due to a sinusoidal input cos(Omega n) is |H(Omega)| cos(Omega n + Theta_H(Omega)). The term -1^n is an input that does not affect the sinusoidal components, so it remains unchanged in the output. The constant term |H(0)| cos(Theta_H(0)) arises from the DC component of the frequency response, which is unaffected by the input's sinusoidal part.*
q18	Determine the Discrete Fourier Series (DFS) coefficients X_i[k] for sampled signals x_i(t) = cos(w_i t + pi/4) with given periods and sampling times: (a) T_1 = 3, Ts_1 = 1; (b) T_2 = 16, Ts_2 = 5; (c) T_3 = 16, Ts_3 = 9.	**Not answerable**	*The DFS coefficients depend on the specific angular frequencies w_i and the sampling parameters Ts_i relative to the signal periods T_i. Without explicit values for w_i, it is impossible to compute the exact DFS coefficients X_i[k]. The general form for DFS coefficients of a cosine signal cos(wt + phi) sampled uniformly is X[k] = (1/(T)) ∑_{n=0}^{N-1} cos(w n + phi) e^(-j 2π k n / N), where N = T / Ts is the number of samples per period. However, w_i is not provided.*
q19	Determine if aliasing occurs for the sampled signals with given periods and sampling times: (a) T_1 = 3, Ts_1 = 1; (b) T_2 = 16, Ts_2 = 5; (c) T_3 = 16, Ts_3 = 9.	**Not answerable**	*Aliasing depends on the signal's frequency content relative to the Nyquist rate (2 * maximum frequency). Since the signal is x_i(t) = cos(w_i t + pi/4), the maximum frequency is w_i/2. The Nyquist rate is 2 * w_i/2 = w_i. Aliasing occurs if the sampling frequency (1/Ts_i) is less than w_i. However, w_i is not provided, so aliasing cannot be determined.*
q20	Find the DFS coefficients of y[n] for discrete-time system y[n] = 0.5 y[n-2] + u[n] - u[n-2] with input u[n] = 1 + 0.5 sin(pi/2 n) + 0.25 cos(pi n).	**Not answerable**	*The system is recursive with a delay term y[n-2], making it impossible to directly compute DFS coefficients without solving the recurrence relation explicitly or knowing initial conditions y[-2], y[-1]. The input contains sinusoidal components, but the recursive nature of the system complicates direct application of DFS formulas without additional information or solving the difference equation step-by-step for each frequency component, which requires initial conditions not provided.*
q21	Determine the magnitude response plot for the discrete-time system y[n] = 0.5 y[n-2] + u[n] - u[n-2] from given options.	**Plot A**	*The system is a second-order difference equation with zero initial conditions. The z-transform analysis shows it is a bandpass filter with a specific magnitude response shape characteristic of such systems, matching Plot A.*
q22	Which expression approximates the given low-pass filter H(z) based on its magnitude response W(Omega)?	**H(z) ~ 1 - W(z)**	*The magnitude response of a low-pass filter is often approximated by subtracting the passband response from 1. Given the description of W(Omega), this corresponds to H(z) ~ 1 - W(z).*

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q23	Determine the entries [Phi]_il of the DFT matrix Phi such that y = Phi b.	**Phi is the DFT matrix where [Phi]_il = (1/sqrt(N)) * exp(-j * 2 * pi * i * l * k / N) for i, l = 0, ..., N-1.**	*The DFT matrix entries are defined by the discrete Fourier transform formula, involving complex exponentials scaled by 1/sqrt(N).*
q24	Show that linearity holds for the DFT transform y = Phi b.	**Linearity holds because the DFT is a linear transformation: Phi(a b) = a Phi(b) for any complex scalar a and vector b.**	*The DFT matrix Phi satisfies the property of linearity for linear transformations, meaning scaling the input vector b by a scalar a scales the output y by the same scalar a.*